REAL ESTATE DATA ANALYSIS PROJECT

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Project Description

This project was created to demonstrate the process of collecting, processing, and analyzing real estate data using SQL, Python, and Excel.

The dataset contains information about properties located in different cities. Each record includes the city name, property type, price, area, and number of bedrooms. The purpose of the project was to organize the data, perform calculations, and identify useful insights from the dataset.

Technologies Used

- MySQL
- Python
- Pandas
- Excel
- GitHub

Database Creation

A database named “realestate” was created in MySQL. A table called “housing” was designed to store information about real estate properties.

The following attributes were included:

- Property ID
- City
- Property Type
- Price
- Area
- Number of rooms

Several sample records were inserted into the database to simulate a small real estate market dataset.

Data Extraction with Python

Python was connected to the MySQL database using SQLAlchemy and Pandas. The data was retrieved from the database and loaded into a Pandas DataFrame for further analysis.

The dataset was successfully imported and checked for accuracy before performing calculations.

Data Analysis

Several analytical tasks were performed using Python.

A new column called “Price per Square Meter” was calculated by dividing the property price by the area of the property.

The analysis focused on identifying:

- Average property prices by city
- Number of properties by property type
- Price per square meter for each property

These calculations provided a better understanding of the real estate market represented in the dataset.

Excel Reporting

After the analysis was completed, the processed dataset was exported to Microsoft Excel.

Excel was used to create summaries and visualize the results of the analysis.

The following findings were obtained:

Average Property Price by City

The average property price was calculated for each city in the dataset. The results showed differences in housing prices between cities and helped identify the most expensive locations.

Number of Properties by Type

The number of properties was counted for each property type, including apartments, villas, and houses. This analysis provided an overview of the distribution of property categories within the dataset.

Price per Square Meter

The calculated price per square meter allowed for a more accurate comparison between properties of different sizes.

Results

The analysis revealed that property prices vary significantly depending on the city and property type.

Properties located in Baku generally had higher prices than properties located in other cities. Villas represented the highest-priced category of properties, while apartments were the most common property type in the dataset.

The calculated price-per-square-meter metric provided additional insight into property value and market trends.

Conclusion

This project demonstrates a complete data analysis workflow using SQL, Python, and Excel.

The project involved database creation, data extraction, data processing, calculation of key metrics, and reporting of analytical results.

The experience gained from this project helped develop practical skills in database management, Python programming, data analysis, and business reporting.

This project serves as an example of how data can be transformed into meaningful information to support decision-making in the real estate sector.